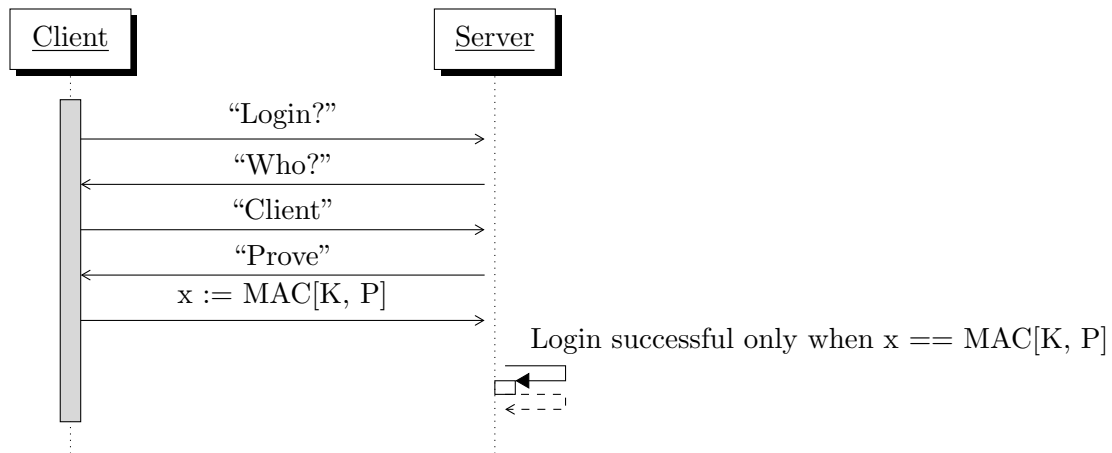


**1.2 (16 points)** Consider the following authentication protocol in which the server authenticates the client. You can assume that the server and client share a symmetric cryptographic key  $\mathbf{K}$ , which they established during the registration phase. In the registration phase, they also establish a very *secure password*  $\mathbf{P}$ . Assume that the server and client both protect the password and key carefully. You can assume the Message Authentication Code (MAC) scheme used in the protocol is secure. For the given authentication protocol, answer the following two questions. **Please consider the channel between the server and client to be insecure, and can be controlled by an active attacker.**



**A (2 + 4 points).** Given the security requirement expected from this protocol is that only the client, who has access to the password  $\mathbf{P}$  and symmetric key  $\mathbf{K}$ , can successfully authenticate itself to the server, is the protocol secure? Show an attack if the protocol is insecure!

**B (10 points).** If the protocol is insecure, please fix the protocol to avoid the attack.